MILITARY SPECIFICATION

TAPE, INSULATING, THERMAL

1. SCOPE AND CLASSIFICATION

1.1 Scope.- This specification covers thermal insulating tape for use as an insulation for piping and tubing at temperatures up to 750°F. It is especially suitable for small piping and tubing where space conditions aboard Naval vessels render awkward the use of molded insulation covering.

1.2 Type.- Thermal insulating tape shall be of the following types, as specified (see 6.1):

Type A - For spiral wrapping.
Type B - For lateral wrapping.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications.- The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

Military Specifications
JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-B-107 - Boxes, Wood, Wirebound (Overseas Type).
JAN-P-125 - Packaging and Packing for Overseas Shipment, Barrier-Materials, Waterproof, Flexible.

Navy Department Specification
General Specifications for Inspection of Material.

(Army.- Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies).

(Navy.- Copies of Military (including Joint Army-Navy and National Military Establishment specifications) and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts Navy Department, Washington 25, D.C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Air Force.- Copies of Military specifications (including Joint Army-Navy and National Military Establishment specifications) may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Marine Corps.- Copies of Military specifications (including Joint Army-Navy and National Military Establishment specifications) may be obtained upon application to the Quartermaster General, Headquarters U.S. Marine Corps, Navy Department, Washington 25, D.C. or the Depot Quartermaster Marine Corps Depot of Supplies, 1100 South Broad Street, Philadelphia 46, Pa. Both the title and identifying number or symbol should be stipulated when requesting copies.)

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2.2 Other publications.- The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Bureau of Supplies and Accounts Publication
Navy Shipment Marking Handbook.

(Copies of the Navy Shipment Marking Handbook should be obtained from sources given for obtaining specifications for the Navy.)

3. REQUIREMENTS

3.1 Material.- Thermal insulating tape shall be composed of a woven asbestos fiber or fibrous-glass felting or silver. The fibrous glass shall be a high grade glass of soft, flexible temper, free from shot, dust, or excessive oil or moisture.

3.2 Jacket.-

3.2.1 Construction.- The jacket shall be woven from asbestos yarn containing not less than 90 percent, by weight, of asbestos fiber having not less than 12 percent, by weight, water of hydration. The jacket may be either tubular woven in one piece or fabricated of asbestos cloth sewed into tubular form.

3.2.2 Weave.- The weave of the jacket shall consist of not less than sixteen strands per inch in the warp and not less than seven strands per inch in the filling. (The jacket may have an open weave on one side consisting of not less than eight strands per inch in the warp and seven strands per inch in the filling.)

3.3 Felting or sliver.-

3.3.1 Construction.- The felting or sliver may be composed of one or more fabricated or felted rovings of either fibrous glass or carded amosite asbestos fibers containing not less than 95 percent by weight, of asbestos. When a single felted roving construction is used, the felting or sliver shall be suitably reinforced and held in shape as a single unit by eight or more loosely spiralled or braided two-ply asbestos or glass-fiber strands. When a fabrication of more than one fibrous glass or asbestos-fiber roving is used to construct the felting or sliver, the rovings shall be held in place and together by a one-over-one-under twill weave of two-ply glass or asbestos-fiber yarns, spaced approximately 7/8 inch apart.

3.3.2 Reinforcing strands.- Plain and wire-inserted reinforcing strands will be allowed, if required, in the construction.

3.3.3 Wire.- The wire used in reinforcing strands shall be of brass, copper, or nickel-copper-alloy, 0.008 inch in diameter, plus or minus 0.001 inch.

3.3.4 Yarn.- The asbestos or glass-fiber yarns shall be two-ply. The asbestos-fiber yarn shall contain not less than 90 percent, by weight, of asbestos fiber having not less than 12 percent, by weight, water of hydration.

3.3.5 Fineness of fibers.- The diameter of individual glass fibers shall average not more than 0.0008 inch. The maximum diameter of any fiber in the test sample shall be not more than 0.0015 inch.

3.3.6 Alkalinity.- The alkalinity of the fibrous glass, expressed as sodium oxide, shall not exceed 0.20 percent.

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3.4 Loss in weight.- Tape in finished form shall not show a loss in weight greater than 15 percent, after subjection to the test described in 4.3.6.

3.5 Tensile strength.- The tensile strength of the asbestos cloth jacket shall be not less than 1,000 pounds per square inch. The asbestos cloth shall not show a loss in tensile strength greater than 50 percent when subjected to the test described in 4.3.5.

3.6 Type A - for spiral wrapping.-

3.6.1 Weave.- The open weave of the jacket shall be not more than 1-3/4 inches wide for 2-inch material and not more than 2-1/4 inches wide for 2-1/2 inch material.

3.6.2 Dimensions.- Thermal insulating tape shall be 2 to 2-1/2 inches wide and 1/4 to 3/8 inch thick, as specified (see 6.1).

3.6.3 Weight.- Thermal insulating tape shall weigh not less than 5 pounds per inch width per 100 linear feet for tape with a fibrous-glass felting or sliver, and not less than 6-1/2 pounds per inch width per 100 linear feet for tape with an asbestos-fiber felting or sliver. The tubular jacket shall be not more than 70 percent, by weight, of the finished tape made with either a fibrous-glass or an asbestos-fiber felting or sliver.

3.7 Type B - for lateral wrapping.-

3.7.1 Weave.- The open weave of the jacket shall be not more than 3-1/4 inches wide.

3.7.2 Dimension.- Thermal insulating tape shall be 5-1/4 inches wide and 3/8 inch thick, unless otherwise specified (see 6.1).

3.7.3 Weight.- Thermal insulating tape shall weigh not less than 25 pounds per 100 linear feet for tape with a fibrous-glass felting or sliver, and not less than 40 pounds per 100 linear feet for tape with an asbestos-fiber felting or sliver. The tubular jacket shall be not more than 55 percent by weight, of the finished tape made with either a fibrous-glass or an asbestos-fiber felting or sliver.

3.8 Workmanship.- The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.- For purpose of sampling, a lot shall consist of all material of one type and size offered for delivery at one time.

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4.2 Inspection (at the place of manufacture).-

4.2.1 Sampling.- Sampling for visual and dimensional inspection shall be in accordance with JAN-STD-105. The Acceptable Quality Level (AQL) shall be "3.2 to 4.4" percent. The single sampling plan for the specified AQL is given in table I. At the option of the Government inspector the equivalent double sampling plan or the sequential sampling plan given in JAN-STD-105 may be used.

Table I - Sampling procedure (AQL 3.2 to 4.4).[1]

Lot size rolls	Sample rolls (single sampling)	Acceptance number	Rejection number
10 to 25	10	0	1
25 to 50	15	1	2
51 to 100	20	2	3
101 to 200	30	3	4
201 to 300	40	3	4

[1] If sample size exceeds inspection lot size, 100 percent inspection shall be used.

4.2.2 If any sample roll is not acceptable it shall be counted with any other defective rolls and compared with the acceptance and rejection numbers of table I to determine whether the entire lot shall be accepted or rejected. Rejected lots shall be re-inspected or replaced by the manufacturer.

4.3 Tests.-

4.3.1 Preparation of sample.- Cut small strips from different parts of the sample.

4.3.2 Hygroscopic moisture.- A known weight of approximately 2 gm is dried for 1 hour at 105°-110° C. in an electric oven, cooled in a desiccator, and weighed; the loss in weight is termed the hygroscopic moisture. This weight, divided by the weight of the sample and multiplied by 100, gives the percentage of hygroscopic moisture in the sample of cloth.

4.3.3 Cotton.- A known weight of approximately 1 gm of the material, dried 1 hour at 105°-110°C., is transferred to a porcelain or platinum boat and inserted in a combustion glass tube in an electric organic combustion furnace; the tube is heated for 1 hour at a low red heat, during which time a current of oxygen gas traverses the tube. This gas is passed through two U-tubes containing calcium chloride to remove moisture and finally into a Vanier or similar absorption bulb containing a strong solution of caustic potash which absorbs the carbon dioxide produced by the burning of the cotton in the sample. Three-elevenths of the increase in weight of the Vanier bulb represents the weight of the carbon in the cotton of the cloth or yarn. This is 44.40 percent of the cotton. Therefore, percentage of cotton equals:

$$\left(\frac{3/11 \times \text{increase in weight of Vanier bulb}}{0.444 \times \text{weight of dried sample}}\right) \times 100$$

This result is reported as the percentage of cotton in dried yarn or cloth. The percentage of asbestos in the dried material equals 100 minus the percentage of cotton. The oxygen gas used should be purified in the usual manner.

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4.3.4 Water of hydration.- Take a known weight of approximately 1 gm of the dried material in a platinum crucible and ignite in an electric furnace at cherry red heat or over a blast lamp to constant weight. The ignited weight of the asbestos, divided by the original weight of the sample and multiplied by 100, gives the percentage of ignited asbestos in the packing. The percentage of ignited asbestos plus the percentage of cotton, subtracted from 100, gives the percentage of water of hydration in the asbestos.

4.3.5 Tensile strength.- Tests shall be made on the three specimens as received and on the three specimens after subjection to soaking heat for 6 hours at 750°F. The jaws of the testing machine shall be set at 10 inches between grips and the machine operated at a speed of 12 inches per minute. The results reported shall be the average of each group of three tests.

4.3.6 Loss in weight.- Compliance with 3.4 shall be determined by subjecting not less than 1 linear foot to soaking heat for 6 hours at approximately 750°F.; the weight before and after heating shall be determined.

4.3.7 Alkalinity.- Weigh a 5 gm (plus or minus 0.01) representative sample of the fibrous glass, and introduce into a 500 mℓ pyrex Erlenmeyer flask. Wet with 5 mℓ of 95 percent ethyl alcohol, and add 400 mℓ of distilled water. Reflux for 4 hours plus or minus 5 minutes using suction. At the end of this period, disconnect the condenser and filter at once through No. 41. Whatman paper supported in a Buechner funnel. Wash the flask and material three times with 25 mℓ portions of hot distilled water. Titrate immediately with 0.02N H_2SO_4, using 6 to 8 drops of 1 percent solution of phenol-red indicator, to the disappearance of the pink color. Run a blank determination of the same amount of distilled water and alcohol and correct for any alkalinity shown. The percentage alkalinity as Na_2O is calculated from the following formula: percent Na_2O=0.0.0124(mℓ s H_2SO_4 used by sample minus mℓ s H_2SO_4 used by blank).

4.3.8 Diameter of fibers.- A specimen shall be taken from sample and the fiber diameter determined microscopically, based on at least five checks from each specimen. If either the maximum diameter of any one fiber thus noted, or the average of sixty such diameters exceed the corresponding limit specified in 3.3.5 the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Tape shall be delivered in rolls containing 25 linear feet. Each roll shall be separately wrapped with a substantial wrapping paper.

5.2 Packing.-

5.2.1 For domestic shipment.- The subject commodity, packaged as specified in 5.1, shall be packed in substantial commercial containers so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

5.2.2 For overseas shipment.- Tape packaged as specified in 5.1 shall be overpacked in cleated plywood, nailed wood, or wirebound boxes conforming to Specifications JAN-P-105, JAN-P-106, and JAN-B-107, respectively. Two-hundred-pound test single wall fiberboard boxes may be used when overpacked for export shipment. Each box shall be lined with a sealed waterproof bag conforming to Specification JAN-P-125. The seams and closures shall have strength and water-resistance equal to that of the body material. The gross weight shall not exceed approximately 125 pounds.

5.2.3 For domestic or overseas shipment.- Where practicable, shipping containers shall be uniform size and shall contain the identical number of rolls of one thickness. Containers shall be designed to fit the contents in a compact manner.

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5.3 Marking.- In addition to any special marking required by the contract or order interior containers and shipping containers shall be marked in accordance with the Navy Shipment Marking Handbook.

6. NOTES

6.1 Ordering data.- Requests, requisitions, schedules and contracts or orders should specify the following:

Title, number, and date of this specification.
Type required (see 1.2).
Dimensions of tape (see 3.6.2 and 3.7.2).

6.2 Thermal insulating tape should be purchased by the linear yard.

6.3 Copies of this specification may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D.C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise, as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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